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Write a program to implement Linear Regression (LR) algorithm in python

CODE :

```
import numpy as np

# Training data

X = np.array([1, 2, 3, 4, 5])

Y = np.array([2, 4, 5, 4, 5])

# Calculate mean

X_mean = np.mean(X)

Y_mean = np.mean(Y)

# Calculate slope (m)

m = np.sum((X - X_mean) * (Y - Y_mean)) / np.sum((X - X_mean) ** 2)

c = Y_mean - m * X_mean

# Linear regression equation: Y = mX + c

print("Linear Regression Equation:")

print("Y =", round(m, 2), "X +", round(c, 2))

Y_pred = m * X + c

print("\nActual Values:", Y)

print("Predicted Values:", np.round(Y_pred, 2))

mse = np.mean((Y - Y_pred) ** 2)

print("\nMean Squared Error:", round(mse, 2))

x_new = 6

prediction = m * x_new + c

print("\nPrediction for X =", x_new, ":", round(prediction, 2))
```

OUTPUT

```
Linear Regression Equation:
Y = 0.6 X + 2.2

Actual Values: [2 4 5 4 5]
ERROR!
Predicted Values: [2.8 3.4 4.  4.6 5.2]

Mean Squared Error: 0.48

Prediction for X = 6 : 5.8

=== Code Execution Successful ===
```